

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 1st Term Examination, 2022
Department of Materials Science and Engineering

MSE 2101
(Crystallography)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY Three** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY Three** questions from this section in **answer script A**)

1. (a) What is lattice and crystal? "The smallest unit having full symmetry of the crystal is called unit cell." Explain the justification of this statement. (10)
(b) Describe the Zinc Blende structure with neat sketch and justify such coordination configuration. (Given, ionic radius of Zn and S are 74 pm and 184 pm respectively) (10)
(c) Briefly explain about crystalline polymer. 67.047% crystallinity of a branched polymer specimen with density of 0.886 g/cm³, totally amorphous density 0.816 g/cm³. What is the perfectly crystalline density of that polymer? (15)
2. (a) How many types of interstitial voids are in different atomic stacking? With necessary diagram calculate the number of tetrahedral and octahedral voids surrounding an atom in closed packed stacking. (12)
(b) Calculate the ideal c/a ratio of HCP structure. (08)
(c) Write down the general formula and explain with necessary diagram of normal spinel and inverse spinel structures. (15)
3. (a) Determine the planar density of (111) plane in FCC unit cell. (12)
(b) Explain the reason behind selecting 4-coordinate system instead of conventional 3-coordinate system for hexagonal crystal system. (11)
(c) Draw the following planes and directions in cubic unit cell (321), (201), ($1\bar{1}1$) and [101], [122], [111]. (12)
4. (a) Prove that, the possibility of five, seven, eight (and so on) fold rotational axes symmetry are not possible in an ideal crystal structure. (10)
(b) With neat sketch show all the symmetry elements that a cubic system contains. (13)
(c) Define primitive and non-primitive unit cell. How does crystal symmetry affect the properties of crystal? (12)

Section B

(Answer **ANY Three** questions from this section in **answer script B**)

5. (a) What do you mean by family of planes? Find out the zone axis of the planes (111) and (110). (12)
- (b) Draw the planar view (with Miller indices) of a BCC crystal perpendicular to the Z-axis. From this planar view, draw a two-dimensional reciprocal lattice and then extend it to a three-dimensional reciprocal unit cell. (15)
- (c) Write a short note on Brillouin Zone. How can you differentiate between Wigner-Seitz cell and Brillouin Zone? (08)
6. (a) What is meant by X-ray diffraction? Differentiate between X-ray diffraction and reflection of visible light. (10)
- (b) Draw neat sketches of Debye-Scherrer film and explain the analysis of crystal structures by the Debye-Scherrer method. (12)
- (c) An X-rays beam have a minimum wavelength of $0.5 \text{ \AA}$ and makes an 60° angle with the surface of the crystal [$d_{001} = 1 \text{ \AA}$]. So, how many orders of reflections are possible and why? (08)
- (d) Write down the geometrical facts of Bragg's law. (05)
7. (a) "Engineering projection is a distance true projection in contrast, stereographic projection is an angle true projection." Explain with necessary figures. (10)
- (b) (i) Draw a stereographic projection of a cubic crystal on (001) plane using Tracing paper and Wulff Net. (08)
- (ii) Draw in the poles of (011) and $(\bar{1}11)$ plane normal and label their traces, then measure the angle between $[\bar{1}11]$ and $[001]$ poles using Tracing Paper and Wulff Net. (17)
8. (a) Discuss the differences between XRD and Electron diffraction. Between Electron and Neutron diffraction which one will you prefer for the determination of magnetic structure of a powdered sample? (15)
- (b) X-rays from a tube undergoes first order reflection at an angle of glancing 12° from the face of a calcite crystal. The grating space of the calcite is $3.04 \times 10^{-8} \text{ cm}$. Calculate the wavelength of the X-rays. (08)
- (c) How does crystallite size and strain affect the peak width of an X-ray diffraction pattern? Explain in brief. (12)

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 1st Term Examination, 2022
Department of Materials Science and Engineering
MSE 2103
(Thermodynamics of Materials)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Explain the generic question addressed by thermodynamics “What happens?” by discussing each labeled points in figure 1(a). (10)

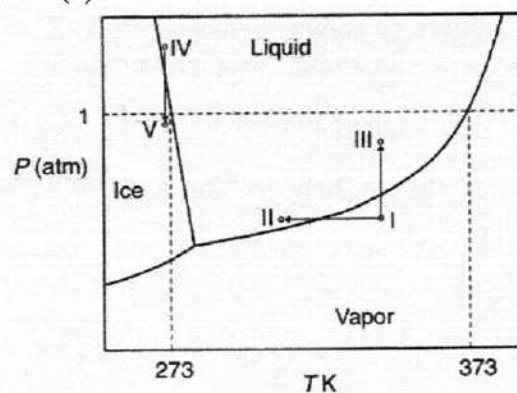


Figure 1(a)

- (b) Classify the following thermodynamic systems (10)
- A eutectic alloy water turbine blade.
 - A porous Styrofoam coffee cup.
 - A can of cold coffee.
 - A Zirconia furnace tube.
 - A glass of ice water.

- (c) Discuss the difference between state function and process variable. The relationship between three variables x , y and z can be represented as $z = 3x^3y + 5x^2y^2$. Find the coefficient relation of this function and comment whether or not these functions are process variables or state function. (15)

2. (a) Define the first and second law of thermodynamics. Derive an expression for the combined statement of first and second law. [Use Table 1]. (10)

Table-01: Thermodynamic State Functions Expressed in Terms of the Independent Variables Temperature and Pressure

$V = V(T, P)$	$dV = V\alpha dT - V\beta dP$
$S = S(T, P)$	$dS = \frac{C_p}{T} dT - V\alpha dP$
$U = U(T, P)$	$dU = (C_p - PV\alpha)dT + V(p\beta - T\alpha)dP$
$H = H(T, P)$	$dH = C_p dT + V(1 - T\alpha)dP$
$F = F(T, P)$	$dF = -(S + PV\alpha)dT + PV\beta dP$
$G = G(T, P)$	$dG = -S dT + V dP$

- (b) Express the change in enthalpy as a function of volume and entropy. (15)
- (c) The combined statement of the first and second laws of thermodynamics evaluates the heat absorbed and mechanical work done on a system with relationship only valid for “reversible processes”. Since reversible processes do not occur in the real world, how is it possible for the combined statement to play an important role in the analysis of practical “irreversible” processes encountered in nature and in technology? (10)

3. (a) What type of thermodynamic variable is heat? Prove your answer. (10)
- (b) A copper rod is heated from 298K to 650K at constant pressure. Calculate the change in entropy, enthalpy and Gibbs free energy for the process. (15)
 $[C_p = 22.6 + 5.6 \times 10^{-3}T \text{ J/mol} - K. S_{298} = 33.14 \text{ J/mol} - K].$
- (c) With the necessary diagram differentiate between steady state and equilibrium state with an example. (10)
4. (a) For a unary two-phase system, derive the conditions for equilibrium using a constrained maximum of entropy. (18)
- (b) One mole of copper initially at 700K and one atmosphere is contained in a thermally insulated jacket. The system is compressed reversibly to a pressure of 10000 atmospheres. Compute the change in temperature for this reversible adiabatic process. (10)
 Given, $V=7.09 \text{ cc/mol}$, $\alpha=51 \times 10^{-6} \text{ K}^{-1}$, $C_p = 22.6 + 5.6 \times 10^{-3}T \text{ J/mol-K}$.
- (c) It is not an overstatement to say that without state functions, thermodynamics would be useless. Discuss this assertion. (07)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Explain the necessity of chemical potential surfaces calculation for the construction of the structure of unary phase diagrams. (10)
- (b) Derive the Clausius-Clapeyron equation which is used most frequently in computing unary phase diagrams. (10)
- (c) Construct the phase diagram for pure silicon using the following informations: (15)
 $T_m = 1685\text{K}$, $T_b = 2873\text{K}$, $\Delta H_m = 50.2 \text{ kJ/mole}$, and $\Delta H_v = 386 \text{ kJ/mole}$.
6. (a) Define fugacity. Derive the Gibbs-Duhem equation for the activity coefficients in a binary system. (15)
- (b) Derive expressions for the partial molar volumes of each components as function of composition if the volume change in a binary solution obeys the relation $V = 2.7X_1X_2^2 \text{ cc/mol}$. (10)
- (c) "For any component in an ideal gas mixture activity coefficient is equal to one"-Prove the statement. (10)
7. (a) Why phase diagrams are necessary? Determine the conditions for equilibrium for multi-component multi-phase systems (isolated system). (20)
- (b) What is degree of freedom? Derive an expression for Gibbs phase rule where the system consists of 'P' phases and 'C' components. (15)

8. (a) Draw the free energy v/s composition diagrams in the indicated temperatures of the phase diagram in figure 8(a). (15)

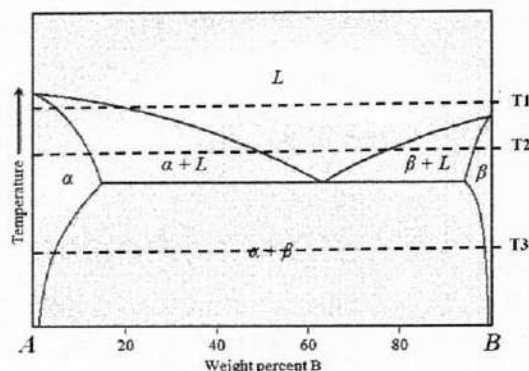


Figure 8(a)

- (b) With the help of Ellingham diagram in Figure 8(b) answer the following questions:

- Explain why the slope of curves in Ellingham diagram changes? And why CO line has negative slope? (10)
- Explain why Titanium can be used to reduce Manganese Oxide but Chromium cannot be used at 800°C ? (10)

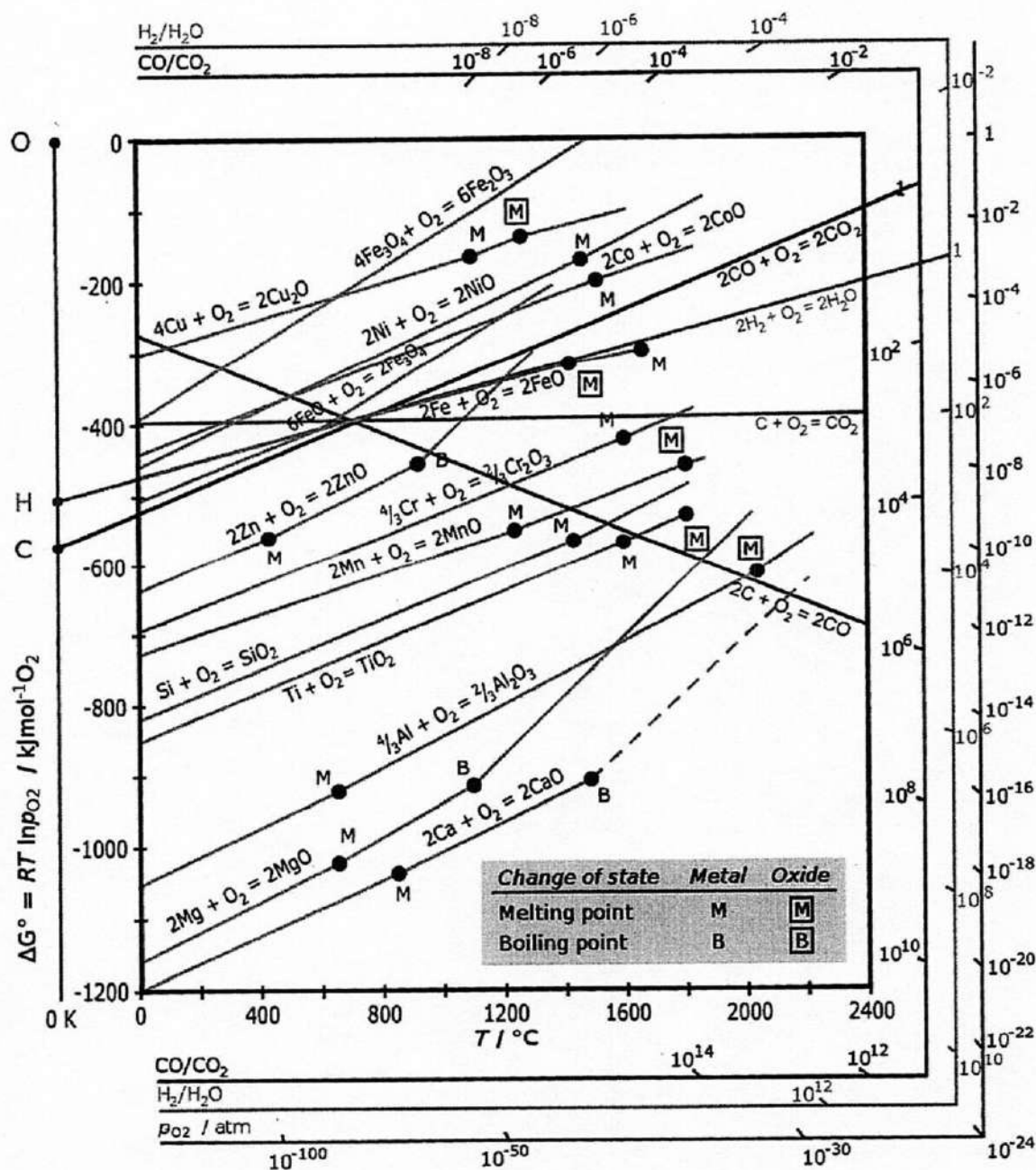


Figure 8(b): The Ellingham Diagram

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 1st Term Examination, 2022
Department of Materials Science and Engineering

Math 2127
(Vector Analysis and Linear Algebra)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) A particle moves along a curve whose parametric equations are $x = 2\sin 3t, y = 2\cos 3t, z = 8t$ at any time $t > 0$. Find (i) magnitude of the velocity and acceleration, (ii) the unit tangent at the point $t = \pi/2$. (12)
- (b) Write the physical significance with mathematical representation of Gradient, Divergence and Curl. (12)
- (c) Evaluate $\nabla \cdot [r \nabla \left(\frac{1}{r^3}\right)]$. (11)
2. (a) Find the directional derivative of the function $\phi = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$ at (1,1,1) in the direction towards the point (2,3,4). (11)
- (b) Find the equations for the tangent plane and normal line to the surface $4z = x^2 - y^2$ at the point (3,1,2). (12)
- (c) A fluid motion is given by $\vec{v} = (y\sin z - \sin x)\hat{i} + (x\sin z + 2yz)\hat{j} + (xy\cos z + y^2)\hat{k}$. Is the motion irrotational? If so, find the velocity potential. (12)
3. (a) Determine the work done in moving a particle in the force field (12)
$$\underline{F} = (2y + 3)\underline{i} + xz\underline{j} + (yz - x)\underline{k}$$
along (i) the straight lines from (0,0,0) to (0,0,1), then to (0,1,1) and then to (2,1,1).
(ii) the straight line joining (0,0,0) and (2,1,1).
- (b) Determine the value of "a" so that the vector $\underline{V} = (x + 3y)\underline{i} + (y - 2z)\underline{j} + (x + az)\underline{k}$ is solenoidal. Examine whether it is a source field or not when $a=1$. (10)
- (c) Evaluate $\iint_S \underline{A} \cdot \underline{n} ds$, where $\underline{A} = 18z\underline{i} - 12\underline{j} + 3y\underline{k}$ and S is the part of the plane $2x+3y+6z=12$ in the first octant. (13)
4. (a) Evaluate $\oint_C (3x^2 - 8y^2)dx + (4y - 6xy)dy$ by using Green's theorem, where C is the curve bounded by $x = 0, y = 0, x + y = 1$. (08)
- (b) If $\iiint_V \nabla \cdot \underline{A} dV = 84\pi$, find $\iint_S \underline{A} \cdot \underline{n} ds$, where $\underline{A} = 4x\underline{i} - 2y^2\underline{j} + z^2\underline{k}$, taken over the region bounded by $x^2 + y^2 = 4, z = 0$ and $z = 3$. (10)
- (c) Define curvilinear coordinates. Prove that a cylindrical coordinate system is orthogonal. (09)
- (d) Derive an expression for $\nabla\phi$ in orthogonal curvilinear coordinates. (08)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Define with example: (i) Elementary matrix, (ii) Augmented matrix of a system, (iii) Periodic matrix and its period. (12)

(b) Define rank of matrix. Find the rank of the matrix A where, $A = \begin{bmatrix} 0 & 1 & -1 & 1 \\ 2 & -1 & 2 & -2 \\ 1 & 0 & -1 & 1 \\ -1 & 1 & -1 & 2 \end{bmatrix}$. (11)

- (c) When a matrix is said to be invertible? If possible, find the inverse of the matrix B by applying elementary transformations where, $B = \begin{bmatrix} -1 & 2 & 3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix}$. (12)

6. (a) Find A^2 by using partitioning where $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{bmatrix}$. (10)

- (b) For which values of a, the following system have (i) no solution, (ii) exactly one solution, (iii) infinitely many solutions? (12)

$$x + 2y - 3z = 4$$

$$3x - y + 5z = 2$$

$$4x + y + (a^2 - 14)z = a + 2.$$

- (c) Define homogeneous and non-homogeneous system of linear equations. Find the trivial and non-trivial solution of the following system of equations (if exists): (13)

$$2x - y - 3z = 0$$

$$-x + 2y - 3z = 0$$

$$x + y + 4z = 0$$

7. (a) Define subspace of a vector space. Determine whether the vectors (1,1,2), (1,0,1) and (2,1,3) span the vector space $\mathbb{R}^3$. (12)

(b) Find a basis for the null space of the matrix A, where $A = \begin{bmatrix} 1 & 4 & 5 & 2 \\ 2 & 1 & 3 & 0 \\ -1 & 3 & 2 & 2 \end{bmatrix}$. (13)

- (c) Express the following quadratic form in the matrix notation $x^T A x$, where A is a symmetric matrix. (10)

$$9x_1^2 - x_2^2 + 4x_3^2 + 5x_1x_2 - 8x_1x_3 + x_2x_3.$$

8. (a) Find the characteristic roots of the following matrix and hence find the corresponding eigen vector for largest root (15)

$$A = \begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}.$$

- (b) State Cayley-Hamilton theorem. Verify it for the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$ and hence find A^{-1} . (10)

- (c) Determine whether the vectors $\underline{V}_1 = (1, -2, 3)$, $\underline{V}_2 = (5, 6, -1)$, and $\underline{V}_3 = (3, 2, 1)$ are linearly independent or not in $\mathbb{R}^3$. (10)

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 1st Term Examination, 2022
Department of Materials Science and Engineering

Hum 2127
(Accounting and Economics)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY Three** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY Three** questions from this section in **answer script A**)

1. (a) State the scope conceptual framework for accounting. (15)
(b) What is accounting cycle? Discuss the steps of accounting cycle. (20)
2. (a) What is trial balance? What are the objectives of preparing a trial balance? (10)
(b) Following are the ledger balances of Argon company, prepare a trial balance as on December 31, 2022. (25)

Account titles	Tk.
Cash	45000
Capital	35000
Supplies	13500
Sales	29000
Supplies expense	2000
Tax payable	20500
Bills receivable	25000
Unearned revenue	14000
Prepaid insurance	2000
Salaries and wages paid	8000
Drawing	3000
Equipment	20000
Accumulated depreciation	6000
Allowance for bad debt	14000

3. Baron company purchased a drilling machine on January 1, 2017. Related data are given below:

Cost =	600000 tk
Salvage value =	50000 tk
Useful life (years) =	5 years
Useful life (hours) =	1650000 (machine hours)

Year	Machine hours used
2017	400000
2018	350000
2019	300000
2020	320000
2021	280000

- Requirements: (a) Show the depreciation schedule of the machine for above five years using straight-line depreciation method. (20)
(b) Do the same using units of activity depreciation method. (15)

4. Emerald company extracted following ledger balances on December 31, 2022:

Accounts	Taka
Purchase return	15000
Owner's equity	300000
Sales	912000
Loans	100000
Account payable	50500
Rent payable	2500
Accumulated depreciation	20000
Cash	85000
Inventory (opening)	170000
Purchases	735000
Misc expense	72000
Sales discount	10000
Interest expense	15000
Account receivable	43000
Buildings	270000

Adjustments: Inventory as on December 31, 2022 is tk. 110000.

Requirements:

- (i) Prepare a statement of comprehensive income for the year ended. (20)
- (ii) Prepare a statement of owner's equity (05)
- (iii) Prepare a statement of financial position as on December 31, 2022. (10)

Section B

(Answer **ANY Three** questions from this section in **answer script B**)

- 5. (a) Define Economics and distinguish between positive and normative economics. (20)
- (b) What is scarcity and choice? Explain the key problems of an economic organization. (15)
- 6. (a) How could you derivate the condition of consumer equilibrium if one's utility function is $U = f(q_1, q_2)$ and budget constraint is $M = p_1q_1 + p_2q_2$? (20)
- (b) How could you separate price effect into substitution effect and income effect? (15)
- 7. (a) Show how in perfectly competitive equilibrium the price of a commodity is equal to its marginal and average cost of production. Illustrate your answer diagrammatically. (20)
- (b) Explain Market structure. (15)
- 8. (a) Distinguish between GNP and GDP. (10)
- (b) Derivate equilibrium of National income & consumption from the following National income model: (15)

$$\begin{aligned}
 Y &= C + I \\
 C &= 100 + 0.2yd \\
 T &= 10 + 0.1y \\
 I &= 50
 \end{aligned}$$

- (c) Discuss the economic consequences of inflation. (10)

ME 2127
(Solid Mechanics)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Explain the following terms: (08)
(i) Hook's law, (ii) Poisson's ratio, (iii) Yield strength and (iv) Rupture strength.
- (b) Show that longitudinal stress is one-half of tangential stress in thin-walled cylindrical pressure vessel. (12)
- (c) Compute the shearing stress in the pin at B for the supported as shown in figure 1(c). The pin diameter is 20 mm. Assume double shear for the pin at B. (15)

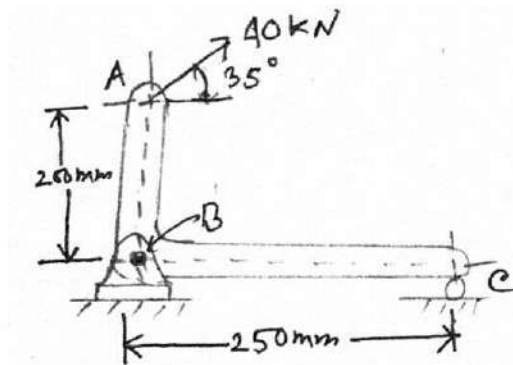


Figure 1(c)

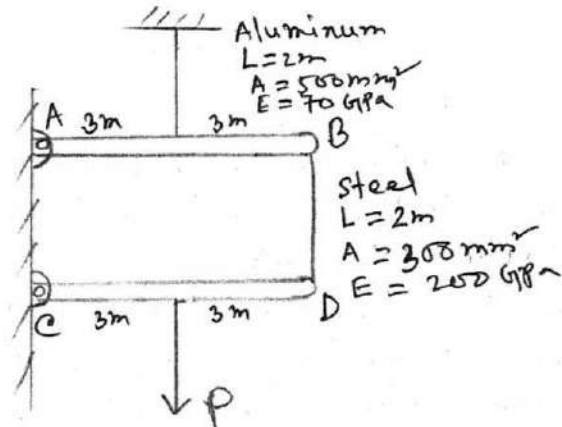


Figure 2(a)

2. (a) The rigid bars AB and CD shown in figure 2(a) are supported by pins at A and C and the two rods. Determine the maximum force P that can be applied as shown if its vertical movement is limited to 5 mm. Neglect the weights of all members. (18)
- (b) Steel railroad rails 10m long are laid with a clearance of 3 mm at a temperature of 15°C. At what temperature will the rails just touch? What stress would be induced in the rails at that temperature if there were no initial clearance? Assume $\alpha = 11.7 \mu\text{m}/(\text{m } ^\circ\text{C})$ and $E = 200 \text{ GPa}$. (17)
3. (a) A compound shaft consisting of a steel segment and an aluminum segment is acted upon by two torques as shown in figure 3(a). Determine the maximum permissible value of T subjected to the following conditions: $\tau_{st} \leq 83 \text{ MPa}$, $\tau_{al} \leq 55 \text{ MPa}$ and the angle of rotation of the free end is limited to 6° for steel. $G = 83 \text{ GPa}$ and for aluminum $G = 28 \text{ GPa}$. (18)

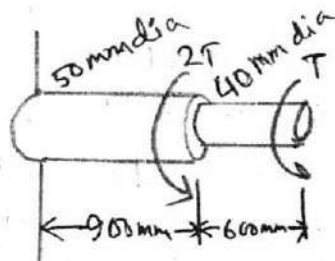


Figure 3(a)

- (b) A load P is supported by two springs arranged in series as shown in figure 3(b). The upper spring has 20 turns of 20 mm diameter wire on a mean diameter of 150 mm. The lower spring consist of 15 turns of 10 mm diameter wire on a mean diameter of 130 mm. Determine the maximum shearing stress in each spring if the total deflection is 80 mm and $G=83$ GPa. (17)

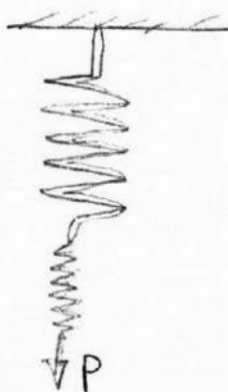


Figure 3(b)

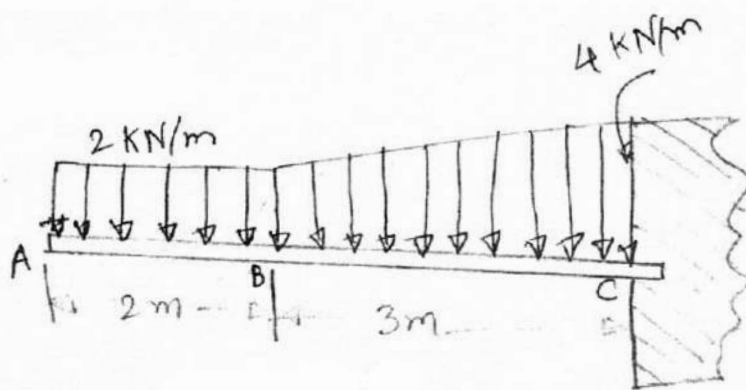


Figure 4(b)

4. (a) What is meant by flexural stress? Show that the maximum flexural stress occurs at the section of maximum bending moment. (17)
- (b) Write shear and moment equations for the cantilever beam carrying loads as shown in figure 4(b). Assume x be the distance measured from the left end of the beam. Also, draw shear and moment diagrams, specifying values at all change of loading positions. (18)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Compute the value of EIS at midspan for the beam loaded as shown in figure 5(a). If $E=10$ GPa, what value of I is required to limit the midspan deflection to $1/360$ of the span. (18)

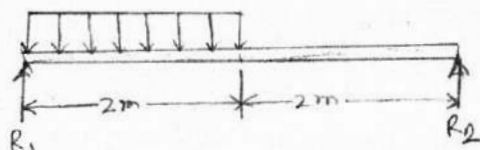


Figure 5(a)

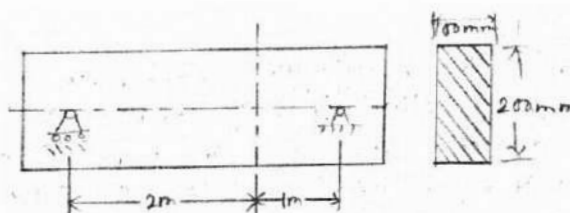


Figure 5(b)

- (b) A wooden beam 100 mm by 200 mm supported as shown in figure 5(b), carries a load P . (17)
What is the largest safe value of P if the maximum stress is not to exceed 10 MPa?
6. (a) Using the method of joints, determine the force in each member of the truss shown in figure 6(a). State whether each member is in tension or compression. (18)

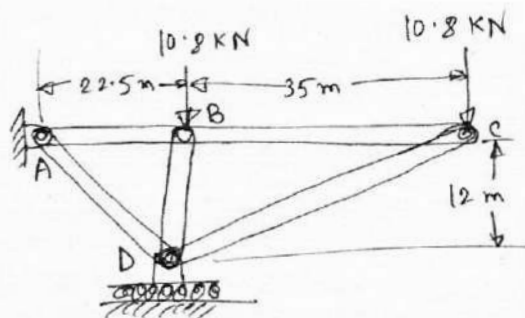


Figure 6(a)

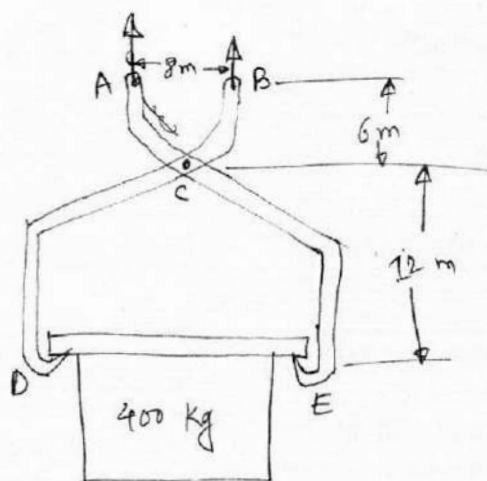


Figure 6(b)

- (b) A cylinder weights 400 Kg and is lifted by a pair of tongs as shown in figure 6(b). Determine the force exerted at D and C on the tong BCD. (17)

7. (a) By direct integration method, determine the centroid of the area shown in figure 7(a). Express the answer in terms of a and h. (17)

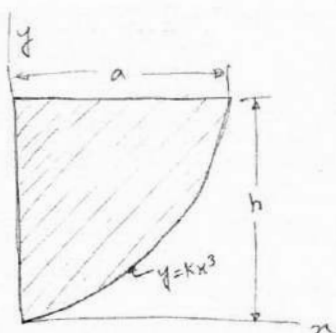


Figure 7(a)

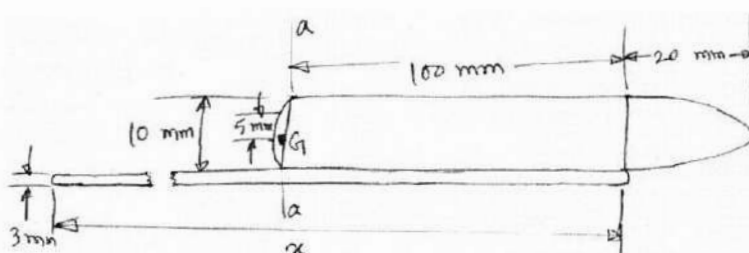


Figure 7(b)

- (b) A toy skyrocket shown in figure 7(b) consists of a solid conical top, $\rho_t = 600 \text{ Kg/m}^3$, a hollow cylinder, $\rho_c = 400 \text{ Kg/m}^3$ and a stick having a circular cross section, $\rho_s = 300 \text{ Kg/m}^3$. Determine the length of the stick, x, so that the center of gravity G of the skyrocket is located along line aa. (18)

8. (a) What are meant by first moment and second moment of an area? What is the significance of radius of gyration? (07)
- (b) Determine by direct integration the moment of inertia of the shaded area with respect to the y-axis. (13)

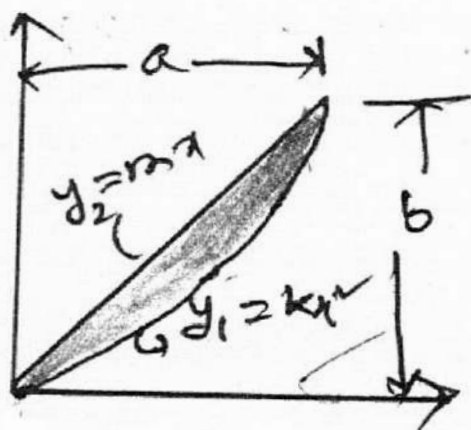


Figure 8(b)

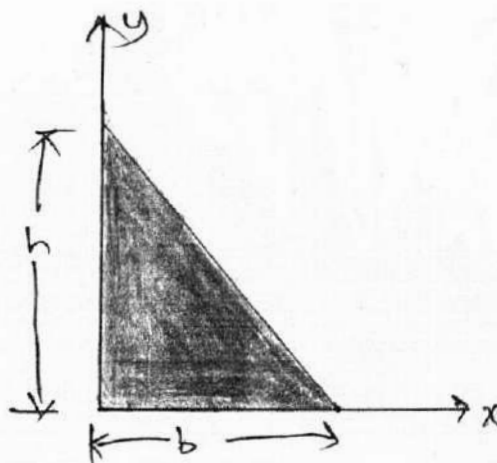


Figure 8(c)

- (c) Determine the product of inertia of the right triangle as shown in figure 8(c), (i) with respect to the x and y axis and (ii) with respect to centroidal axes parallel to the x and y axis. (15)